

LLM Inference Serving

Foundations, systems, and the case for anticipatory control

KV caches, vLLM, LMCache, distributed serving, and agent workloads

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Abstract

An LLM service converts a stream of variable-length, stateful computations into latency-constrained responses. Efficient matrix multiplication is necessary, but so are memory allocation, batching, cache reuse, data movement, routing, and admission control. Agent workloads add a further complication: a session can stop generating tokens while a tool runs, yet return with a large reusable context. The best immediate request decision need not be the best session or fleet decision.

This paper develops the serving stack from first principles, derives the main memory, compute, bandwidth, and queueing relationships, and maps major research and open-source systems onto the problems they solve. It covers paged attention, continuous batching, prefix reuse, cache tiers, prefill/decode disaggregation, parallelism, compression, and program-aware scheduling. It then defines ATFM's potential contribution precisely: using observations available before an agent's next request to estimate future demand and inform bounded control actions. Existing systems already address substantial parts of this problem. The paper therefore gives both the case for ATFM and conditions under which it adds no value, with a comparison and evaluation plan that can falsify the hypothesis. All numerical illustrations are derived examples unless explicitly identified as repository measurements. No new GPU serving benchmark is claimed.

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1 The problem the serving stack actually solves

A model checkpoint describes a computation. A serving system must decide when, where, and with what retained state to perform that computation for many users at once. Two prompts of the same length can have very different service costs: one may reuse a cached prefix on an idle GPU; another may need a full prefill, a remote transfer, or admission behind a long queue. A fleet with fast kernels can still deliver poor latency if it repeatedly recomputes old contexts or routes requests to an overloaded cache-rich worker.

An agent makes the problem more temporal. A coding agent may generate a command, wait for a test suite, append its output, and ask the model again. During the wait, the GPU sees no active decode for that session. Nevertheless, retaining some state may save work when the tool finishes. Retaining everything can also prevent other sessions from making progress. The decision requires the cost of keeping state, the cost of losing it, and the likely time until reuse.

1.1 A map of responsibilities

Figure 1 distinguishes six layers. The application determines semantics and dependencies. A gateway enforces request limits and routes work. An inference engine schedules token computation and manages active state. Kernels implement tensor operations. A cache service extends where reusable state can live. A tool runtime executes non-LLM work. A single product can span several layers; the boundaries describe responsibilities, not mandatory processes.

Table 1: Questions that belong to different layers.

Layer	Main question	Typical cost of a wrong decision
Kernel	How do these tensors execute efficiently?	Extra device IO or compute
Engine	Which token work fits this iteration?	Stalls, preemption, wasted capacity
Cache	Which reusable state should exist where?	Recompute, transfer, occupied memory
Router	Which worker should receive this request?	Queueing or lost locality
Admission	Should this work enter now?	Overload or unnecessary delay
Agent control	What is likely to arrive after a tool?	Missed preparation or wasted reservation

1.2 How to read this paper

Readers new to serving should start with the equations and worked memory example in Section 2, then the engine and cache sections. Readers already operating vLLM can focus on cache movement, agent scheduling, and the ATFM assessment. The appendix provides notation, terminology, a diagnosis guide, and a reading sequence. Equations are models with stated assumptions, not universal performance predictors.

The survey is broad but deliberately bounded. It emphasizes autoregressive transformer inference, open serving infrastructure, and systems relevant to agent traffic. Training, model-quality leaderboards, mobile inference, and every proprietary platform are outside scope. “State of the art” means the relevant research frontier and actively documented mechanisms, not one universally fastest system. Published comparisons differ in hardware, model, workload, cache warmness, implementation generation, and latency constraints. Their headline speedups cannot be put in one league table.

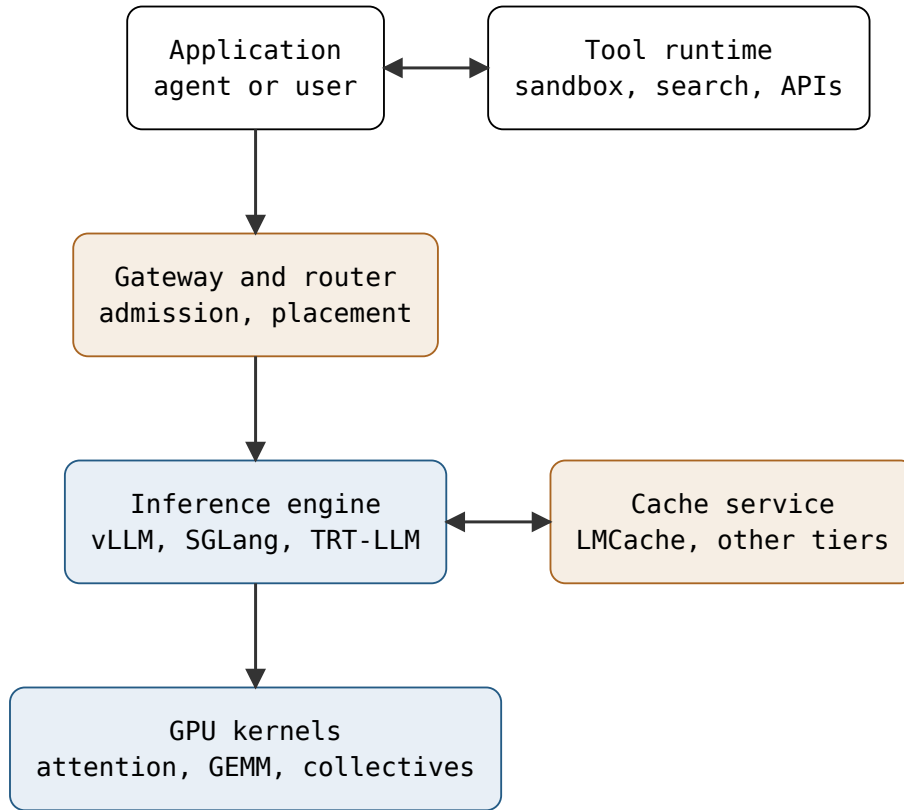


Figure 1: A conceptual serving stack. Arrows denote request or data exchange, not a mandated implementation. ATFM primarily concerns the information and policy between applications and serving; it does not replace GPU kernels.

1.3 Three kinds of evidence

Established mechanism means a cited paper or implementation documents how a technique works. **Analytical illustration** means a derivation or a synthetic example in this paper, with explicit numbers and assumptions. **ATFM evidence** means a result recorded in this repository. A source’s reported benchmark is evidence for that source’s configuration, not an ATFM result. Recent preprints are marked as such; living documentation has an access date. We use primary papers and project documentation, and avoid reproducing vendor peak-throughput rankings.

The central hypothesis is intentionally narrower than “agents need better serving”: conditional information about a paused session can improve a future resource decision enough to pay for obtaining and acting on that information. A forecast with lower statistical error is not sufficient. The decision must change, complete before it matters, and improve a user-relevant outcome.

2 From attention to compute, memory, and latency

2.1 What is stored in a KV cache?

For a single attention head, let a token’s representation be projected into a query q , key k , and value v . Given a prefix of length T , causal attention for its newest token has the form

$$o_T = \text{softmax}\left(\frac{q_T K_{1:T}^\top}{\sqrt{d_h}}\right) V_{1:T}. \quad (1)$$

The keys determine how strongly earlier positions are attended to; values provide the content combined by those weights. This is the standard scaled-dot-product attention construction [1]. The cache stores previously computed keys and values at the required layers. It is neither the model’s learned weights nor a cache of finished textual answers.

For a causal model with an unchanged prefix, earlier hidden states do not depend on future tokens. Their keys and values therefore need not be recomputed when one token is appended. The new query must still attend to the appropriate old state. Caching removes repeated prefix construction; it does not make reading or attending over the prefix free. Queries normally need not be retained for future autoregressive steps.

A cache entry has a computational identity. Matching text alone is insufficient if model weights, adapters, tokenization, positional treatment, or multimodal inputs differ. Reusing state under the wrong identity can silently change the answer. Likewise, the KV for a passage following one prefix generally differs from its KV following a different prefix. Exact prefix reuse and arbitrary passage reuse are distinct mechanisms.

2.2 Prefill and decode are different workloads

During *prefill*, the model processes known prompt tokens. Many token projections and feed-forward operations can execute together as matrix-matrix operations. During ordinary *decode*, the next token is unknown until the previous step completes. Each active sequence contributes roughly one new token to an iteration, although speculative methods alter this unit of work.

Let L be layers, T prompt tokens, d hidden width, and $H_q d_h = d$ for an ordinary attention configuration. Ignoring architecture-dependent constants, a dense model’s prefill work has the form

$$F_{\text{prefill}} = \Theta(LT d^2) + \Theta(LT^2 d). \quad (2)$$

The first term collects projections and feed-forward computation, assuming its intermediate width scales with d ; the second is dense attention. A cached single-token decode step at context length t has work

$$F_{\text{step}} = \Theta(Ld^2) + \Theta(Ltd). \quad (3)$$

For U new output tokens after a T -token prompt, the attention contribution sums as

$$\sum_{u=0}^{U-1} \Theta(L(T+u)d) = \Theta\left(Ld\left(UT + \frac{U(U-1)}{2}\right)\right). \quad (4)$$

These expressions explain why a single cached decode step is linear in context length while a long generation still accumulates a quadratic output-length term. A deliberately naive implementation that reruns full dense attention over the entire growing prefix at every step can accumulate cubic work in final sequence length. That is not how a normal cached inference engine should run. They are arithmetic counts, not wall-clock predictions: batching, memory traffic, collective communication, and kernel utilization matter.

FlashAttention tiles exact attention to reduce traffic to high-bandwidth memory and avoid materializing the full attention matrix there. It does not turn dense attention into a linear-arithmetic algorithm [4]. FlashAttention-3 uses hardware-oriented overlap and low-precision techniques [5]. FlashInfer focuses on flexible, efficient inference kernels across cache layouts and dynamic serving patterns [6]. These optimizations are complementary to scheduling: a better admission policy cannot recover all the performance lost by an unsuitable attention backend.

2.3 The cache-size equation

For a uniform full-attention model, an unsharded sequence’s KV payload is

$$M_{KV}(T) = 2LH_{kv}d_hd_bT, \quad m_{\text{token}} = 2LH_{kv}d_hd_b, \quad (5)$$

where H_{kv} is the number of key/value heads, b bytes per stored scalar, and the leading two counts keys and values. This excludes allocator metadata, quantization scales, temporary buffers, and replication. With B independent sequences of lengths T_i , payload is $m_{\text{token}} \sum_i T_i$ before sharing or architectural exceptions.

Multi-head attention (MHA) uses separate KV heads for query heads. Multi-query attention (MQA) shares one KV head across queries [2]. Grouped-query attention (GQA) sits between them, with several query heads sharing a KV head [3]. Reducing H_{kv} changes model architecture and its stored representation; it is not merely a runtime eviction policy.

Worked example. Take an illustrative $L = 32$, $H_{kv} = 8$, $d_h = 128$, and $b = 2$. Then

$$m_{\text{token}} = 2 \cdot 32 \cdot 8 \cdot 128 \cdot 2 = 131,072 \text{ bytes} = 128 \text{ KiB}, \quad (6)$$

$$M_{KV}(8192) = 1 \text{ GiB}. \quad (7)$$

Sixty-four independent 8192-token contexts consume 64 GiB of payload before weights and other allocations. A hypothetical 80 GiB device with 20 GiB of weights and 8 GiB reserved for runtime overhead leaves 52 GiB, so only 52 such contexts fit under this simplified arithmetic. This is a memory upper bound, not a recommended concurrency or measured GPU capacity. Decode throughput and latency may impose a lower limit.

2.4 Where the simple formula stops applying

Multi-head latent attention, introduced in DeepSeek-V2, stores a compressed latent representation rather than the ordinary full per-head KV tensors [28]. Its memory formula must include that representation and any positional state; substituting an arbitrary small H_{kv} into Equation 5 is not a general MLA estimator. Sliding-window layers retain a bounded attention window; hybrid models may mix full attention, local attention, and recurrent state. vLLM’s hybrid cache manager explicitly handles different layer groups and their cache-hit rules [11].

Tensor parallelism can shard cache heads, but heads or other state may be replicated depending on the model and parallel degree. Pipeline parallelism partitions layers. Context parallelism partitions sequence work/state and introduces communication. Always distinguish total cluster bytes from bytes on the busiest rank. Dividing a global KV estimate by GPU count is valid only if the actual layout shards it evenly. A controller should consume measured per-worker capacity and layout metadata rather than infer them from a model’s parameter count.

2.5 The roofline view: what limits one iteration?

A useful lower-bound model is

$$t_{\text{iter}} \gtrsim \max \left(\frac{F}{P_{\text{eff}}}, \frac{D_{\text{HBM}}}{B_{\text{HBM,eff}}}, \frac{D_{\text{link}}}{B_{\text{link,eff}}} \right). \quad (8)$$

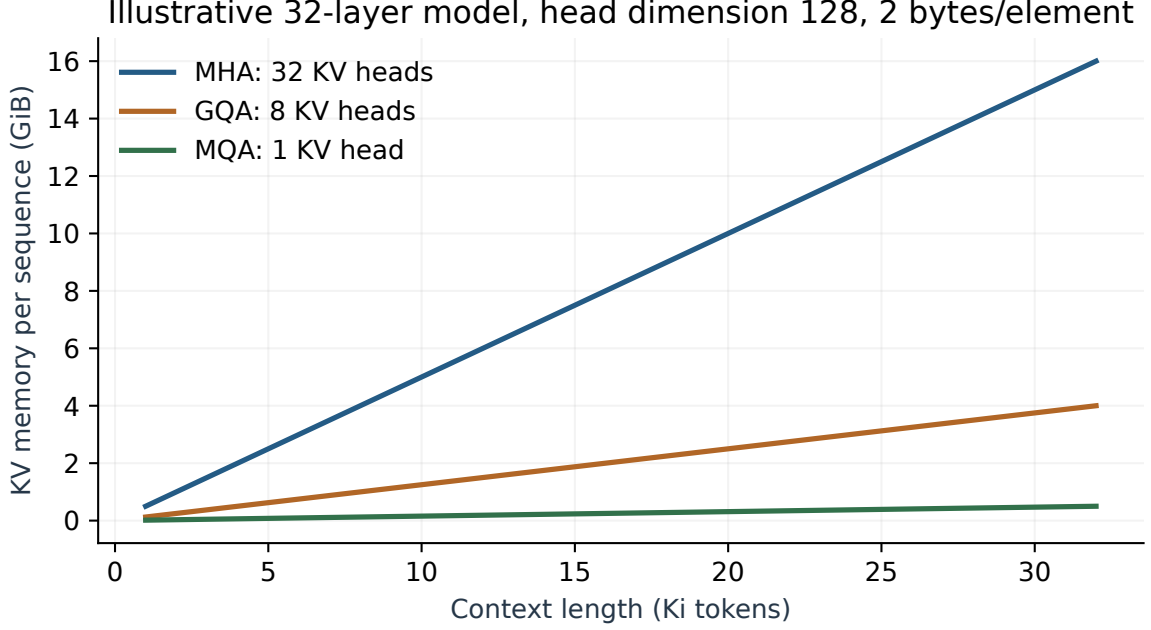


Figure 2: Derived KV payload for an illustrative uniform model. Changing the number of KV heads changes the slope; paging changes allocation waste rather than the payload slope. No hardware was benchmarked.

F is floating-point work; D terms are transferred bytes; P and B are achievable rates for this workload, not device marketing peaks. Real latency also includes launches, synchronization, queueing, and portions that cannot overlap. Adding all three ratios overestimates fully overlapped work; taking only their maximum underestimates serial dependencies. The execution trace determines which is appropriate.

Small-batch dense decode often pays heavily to read weights and KV for few new tokens. Batching amortizes weight traffic across more tokens, but adds KV reads and can increase per-iteration latency. Large prefill can exploit compute well, yet long-context attention or distributed communication may become limiting. “Prefill is compute-bound and decode is memory-bound” is a useful first hypothesis, not a property true for every shape, model, or device.

Arithmetic intensity is $I = F/D_{\text{HBM}}$. In a simplified roofline, achievable FLOPs/s is bounded by $\min(P_{\text{eff}}, IB_{\text{HBM,eff}})$. Optimizing Python dispatch will have little effect while a long GPU kernel is dominant and well fed. It can matter enormously when short iterations wait for CPU scheduling or when a control service saturates a core. These are different critical paths and require different measurements.

2.6 Latency metrics and useful throughput

Let a_i be request arrival, f_i first-token delivery, z_i last-token delivery, and U_i the number of output tokens. Then

$$\text{TTFT}_i = f_i - a_i, \quad (9)$$

$$\text{TPOT}_i = \frac{z_i - f_i}{U_i - 1} \quad (U_i > 1). \quad (10)$$

Individual inter-token latencies (ITLs) are adjacent delivery gaps. TPOT is an average over those gaps and can hide a single long stall. TTFT includes queueing, frontend work, cache retrieval, prefill, and delivery; it is not a direct prefill kernel timer. Client-observed latency also includes transport and buffering.

For horizon H , define request goodput under a chosen joint SLO as

$$G = \frac{1}{H} \sum_i \mathbf{1}_{\{\text{success}_i, \text{TTFT}_i \leq \tau_f, \text{TPOT}_i \leq \tau_o\}}. \quad (11)$$

Report the definition of success, errors, offered load, and rejected requests. A system that rejects difficult work can appear to have excellent latency while serving few users. Token throughput alone can favor longer outputs or a changed workload mix. For agents, whole-job completion time and task success are needed in addition to per-call metrics.

3 Inside an inference engine

3.1 From static batches to iteration-level scheduling

In a static batch, requests can remain tied together until the longest member finishes. A request producing ten tokens then wastes potential participation while another produces a thousand. Iteration-level scheduling chooses the active set again between model iterations, allowing finished requests to leave and new work to enter. Orca established this design for generative-model serving [7]. “Continuous batching” is the common operational term.

Consider three requests with output lengths 2, 5, and 3. In a deliberately simplified model where a three-slot iteration has constant cost, a fixed batch runs five iterations for ten useful token updates out of fifteen possible slots. A dynamic scheduler can fill vacated slots with new requests. This example demonstrates available scheduling opportunity, not a 15/10 speedup: real iteration time changes with batch composition, prompt work, and cache length. The scheduler also needs free state capacity before admitting work.

The engine’s queue is not interchangeable with the external gateway queue. Once admitted, a request may consume KV blocks and compete for token budget. Holding a request outside the engine avoids some of that pressure but adds waiting before execution. A useful gateway admission policy must understand what the engine can already do, otherwise it can reduce batching and leave the GPU idle.

3.2 PagedAttention and the vLLM memory model

The foundational vLLM design maps a sequence’s logical KV blocks to physical blocks allocated as needed. Blocks do not have to be adjacent in GPU memory. This reduces large reservations and external fragmentation, and permits shared blocks with reference counting and copy-on-write behavior [8]. Paging does not compress the tensors in Equation 5.

For a block capacity of p tokens, sequence i with T_i tokens requires $\lceil T_i/p \rceil$ blocks. Its unused tail is

$$w_i = p \lceil T_i/p \rceil - T_i, \quad 0 \leq w_i < p. \quad (12)$$

Ignoring shared blocks, tail waste is less than Bp token slots for B sequences. Larger blocks reduce some metadata and indexing overheads but coarsen allocation and reuse; smaller blocks can reduce unused tails while increasing management work. These are design tradeoffs, not a universal best block size. The appropriate choice also depends on the attention backend and model’s supported layouts.

vLLM’s versioned prefix-cache design identifies full blocks using a chain of hashes over prior-prefix identity, current tokens, and relevant extra identity fields. A new request can reuse a matching prefix; reference counts and a free block structure distinguish currently used from evictable blocks [10]. A prefix can be logically cached yet be evicted when space is needed. An active reference protects state required by a current computation; retaining a finished session’s state is a separate policy problem.

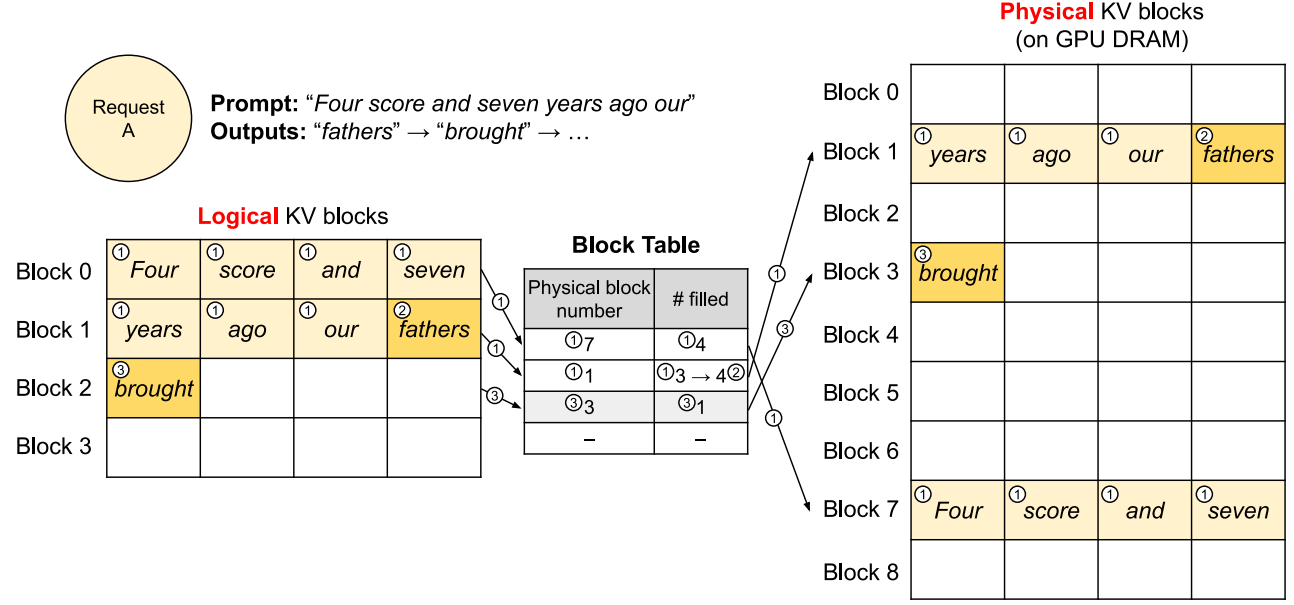


Figure 3: vLLM’s original block-table translation diagram: logical token blocks map to noncontiguous physical KV blocks. Reproduced from Kwon et al. [8], Figure 6, arXiv:2309.06180v1, CC BY 4.0; scaled and format-converted only.

Walkthrough: two related prompts. Suppose prompts are “system + repository + task A” and “system + repository + task B”. If tokenization and all computational identity match through the repository portion, those prefix blocks can be reused. If the first differing token is inside a block, a block-level implementation’s reuse boundary may be before that token. Hash lookup is not a semantic similarity search. Changing a system instruction near the front can invalidate later exact-prefix reuse even when most natural-language content looks similar.

3.3 vLLM V1: token budgets and host overhead

The V1 guide describes a unified scheduler assigning token counts per request, rather than entirely separate scheduling abstractions for prompt and output phases. This supports composing chunked prefill, prefix caching, and speculative work within a token-budget model [9]. The abstraction matters because the engine can control work more finely than a gateway that only counts requests. An 80-token prompt and an 80,000-token prompt are not equal demand.

A conceptual budget is

$$\sum_{i \in \mathcal{A}} n_i^{\text{scheduled}} \leq K, \quad M_{\text{active}} + M_{\text{reserved}} \leq M_{\text{KV, budget}}. \quad (13)$$

A token count is only a proxy for work: decode at long context has a different cost from a short-context token, and full, local, and latent attention differ. The hardware ultimately constrains time and memory, not abstract token units.

The host still tokenizes, validates requests, maintains scheduling metadata, prepares tensors, and processes outputs. A model server can use Python at the orchestration layer while kernels and collectives execute in compiled code. The relevant question is whether Python lies on the measured critical path. Moving every component to Rust does not change the amount of KV data read by a GPU or transferred over a link. Removing repeated scans, redundant RPCs, or avoidable serialization can matter before a language change does.

3.4 Chunked prefill and interference

A large prefill inserted among active decodes can delay their next tokens. Chunking divides prompt work across iterations, placing a bound on how much prefill competes with decode at once. Sarathi-Serve develops scheduling around chunked prefills to manage this throughput/latency tension [17].

Let a chunk have c tokens and estimated execution time $t_p(c, B, T)$ under the current decode set. A conceptual ITL budget imposes

$$t_d(B, T) + t_p(c, B, T) + t_{\text{other}} \leq \tau_{\text{ITL}}. \quad (14)$$

This additive approximation is conservative when work overlaps and optimistic when it ignores interference. Smaller c creates more scheduling boundaries and may hurt compute efficiency. Larger c can improve prompt throughput while worsening decode tail gaps. Adaptive chunking requires measured response surfaces, not simply choosing the smallest chunk.

3.5 Other engine designs and why comparison matters

SGLang. Its original work combines structured LLM-program execution with RadixAttention, representing reusable prefixes in a radix-tree-oriented cache [13]. This is an alternative organization of prefix reuse, not an absence of memory constraints. Its current HiCache design extends reuse beyond GPU memory; Section 4 separates that layer from the original engine mechanism.

TensorRT-LLM. The documented KV subsystem includes block pools, reuse, secondary-memory offload, and prioritized eviction. Retention can attach priorities to token ranges [15]. Therefore a blanket statement that “engines expose no retention controls” is incorrect. ATFM must discover which control is available in the deployed backend and compare against that engine’s best native retention policy.

vAttention. The memory-management paper with this name keeps the virtual address layout contiguous while mapping physical GPU memory on demand using CUDA virtual-memory APIs [16]. It addresses fragmentation without requiring the same non-contiguous virtual layout as PagedAttention. This paper is distinct from a later sparse-attention project with the same name. The broader lesson is that block paging is one way to manage growing state, not a mathematical prerequisite for efficient inference.

3.6 Speculation, quantization, and parallel execution

Speculative decoding generates candidate tokens with a cheaper draft process and verifies them using the target model. With the appropriate acceptance and correction algorithm it can preserve the target sampling distribution [29]. A simple throughput comparison is

$$t_{\text{effective token}} \approx \frac{t_{\text{draft}}(k) + t_{\text{verify}}(k) + t_{\text{control}}}{\mathbb{E}[A(k)]}, \quad (15)$$

where $A(k)$ counts emitted tokens per round under the exact implementation, including any additional verified token. Acceptance, draft overhead, available batching, and memory use determine whether the method helps. A request-level capacity estimator must not assume one target forward pass always equals one emitted token.

Weight quantization reduces model storage/traffic; KV quantization reduces sequence-state traffic. They alter different terms in the budget. KIVI studies asymmetric low-bit quantization of keys and values [27]; low-bit cache storage requires numerical-quality evaluation, scale metadata, and suitable kernels. A factor-of-eight reduction in scalar bit width does not imply an end-to-end factor-of-eight memory or latency reduction.

Parallelism redistributes work rather than removing it. Tensor parallelism splits operations and adds collectives; pipeline parallelism splits layers and can suffer bubbles; data parallelism creates independent model replicas and

needs routing; context parallelism spreads long sequence work; expert parallelism routes mixture-of-experts tokens among expert shards. A fleet may combine them. A cache-local routing decision that ignores collective topology can be worse than a small cache miss on a better-connected worker group.

Practical consequence. Before adding a higher-level controller, establish a tuned engine baseline with continuous batching, appropriate kernels, realistic token budgets, and the native cache/reuse features supported by the chosen model. ATFM should be evaluated on top of that baseline, not credited for optimizations already available in it.

4 Cache hierarchies, movement, and distributed serving

4.1 A cache hit is a location-dependent event

A prefix can be present on the destination GPU, retained in host memory, stored on local SSD, or available at another machine. These cases all avoid some recomputation but have different latency and resource costs. A remote hit may be slower than local recomputation. A GPU hit may still wait behind another request. A cache metadata hit is not proof that the tensors are ready for an attention kernel.

For cache payload M , a useful restore model is

$$t_{\text{restore}} = t_{\text{lookup}} + t_{\text{queue}} + t_{\text{setup}} + \frac{M}{B_{\text{effective}}} + t_{\text{layout}} + t_{\text{sync}}. \quad (16)$$

The terms include object lookup, waiting for IO resources, transfer setup, effective bandwidth, tensor layout conversion, and synchronization. Multiple hops may overlap or serialize. Bottleneck link bandwidth is an optimistic starting point, not necessarily the end-to-end rate. A distributed service must also reserve destination capacity before advertising completed placement.

Restore versus recompute. A local decision compares completion costs, not merely existence:

$$t_{\text{restore}} + \Delta t_{\text{interference}} < t_{\text{recompute}} + \Delta t_{\text{alternative queue}}. \quad (17)$$

The extra terms matter when restores congest a link or recomputation occupies a GPU that could serve other requests. For illustration, transferring 1 GiB at 25 GiB/s takes 40 ms before overhead; at 8 GiB/s it takes 125 ms. If fixed overhead is 1 ms and recomputation costs 150 ms, both may help in isolation. At 2 GiB, the slower path is already 251 ms. These rates and the recomputation cost are assumptions, not device specifications or measurements.

4.2 LMCache: externalizing reusable inference state

LMCache provides a cache layer that extracts, retains, retrieves, and exchanges KV state around inference engines. Its technical report describes reuse across requests and engines, prefill/decode transfers, batched movement, IO/compute pipelining, and control APIs [21]. It is not an LLM, tokenizer, or replacement for the engine’s attention scheduler. External cache capacity is useful only if the correct state can be retrieved at a cost lower than the alternative.

The current multiprocess architecture documentation places cache management in a separate server. A storage manager composes local memory management, secondary storage adapters, and asynchronous store, prefetch, and eviction controllers. The documented local manager can be backed by CPU DRAM or a GPU-direct-storage NVMe slab [22]. Therefore describing every LMCache deployment as “a Python dictionary of CPU tensors

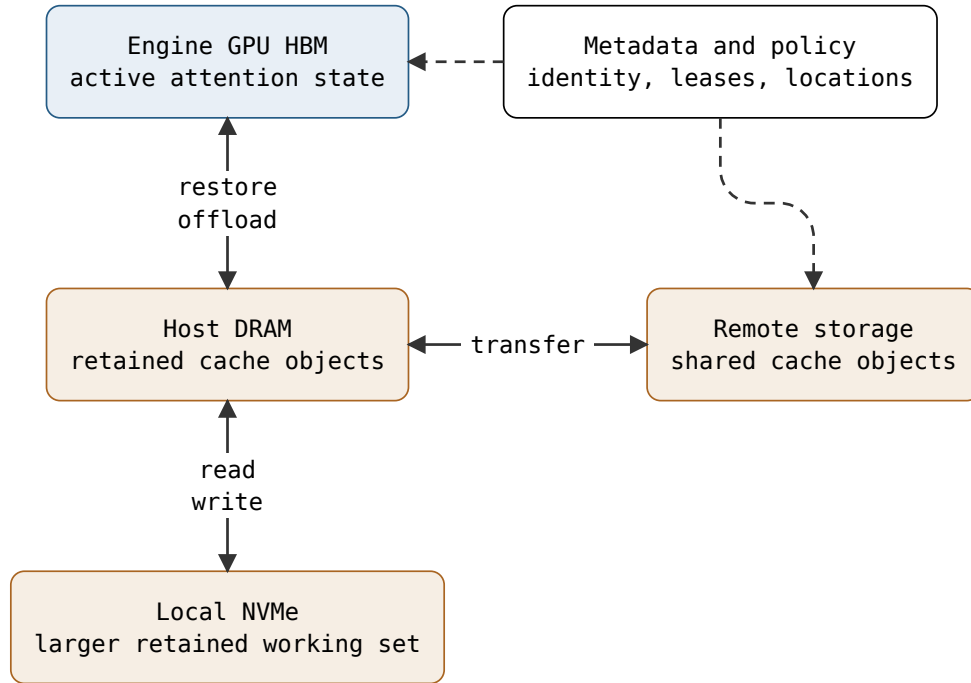


Figure 4: A conceptual KV hierarchy. Storage tiers hold reusable state; the inference engine owns the state required for active attention. Metadata/control paths are dashed. Product-specific tier names are not universal: SGLang HiCache and LMCACHE use different L1/L2 terminology.

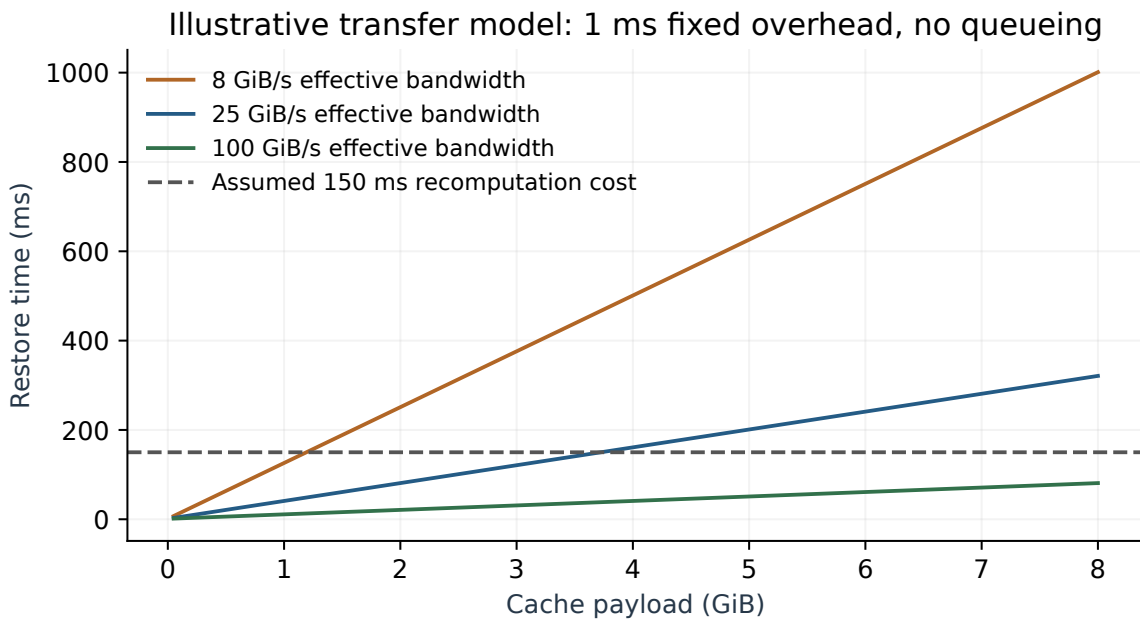


Figure 5: Analytical restore-time curves from Equation 16 with all terms except setup and transfer set to zero. The horizontal line is an assumed recompute cost. Queueing or layout conversion moves the crossover.

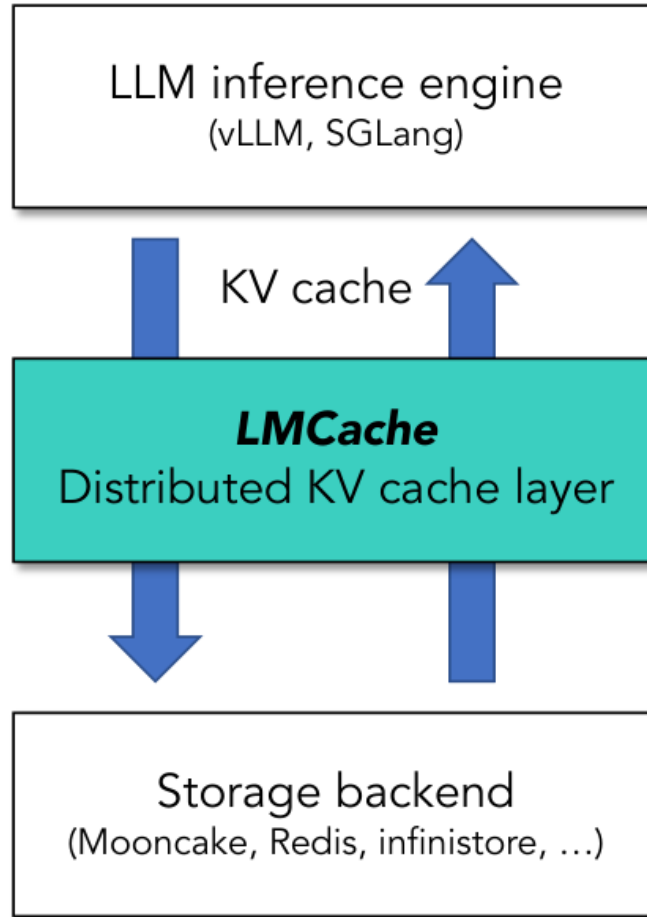


Figure 6: LMCache’s technical-report architecture connects inference engines to storage and transport backends. Reproduced from Liu et al. [21], Figure 5, arXiv:2510.09665v2, CC BY 4.0; scaled only. This depicts the report’s architecture; the current multiprocess implementation is discussed below.

inside vLLM” is both incomplete and stale. A deployment must specify its mode, cache ownership, transport, storage backend, and connector versions.

A conceptual read path is: identify a reusable prefix; discover available objects; obtain valid access/retention while transferring them; populate the engine’s expected representation; wait for readiness at the required layer or batch boundary; compute the unmatched suffix. A write path must decide when and how much completed state to retain. These are generic correctness steps, not a claim that a particular LMCache release exposes one API per step.

The integration contract that ATFM must verify. The legacy in-process controller page is now explicitly deprecated in favor of multiprocess mode [23]. Its pin API identifies tokens, an instance, and a storage location, returning an operation event identifier; the documented example pins a CPU backend [24]. That does not establish an engine-HBM reservation, nor does an accepted asynchronous operation prove transfer completion. ATFM’s current HTTP adapter uses older controller-shaped operations and configurable location strings. Real-worker compatibility is an open integration task, not something its mock HTTP tests settle.

To make a placement claim defensible, record the engine and connector versions, resolve session/prompt identity to exact token/cache identity, verify the supported location, observe operation completion, and check subsequent reuse. If only a host copy is protected, the expected benefit is avoiding a deeper reload or recomputation, not

guaranteed zero-copy GPU reuse. This distinction belongs in both experiments and user-facing descriptions.

4.3 Other cache hierarchies and advanced reuse

SGLang HiCache organizes GPU, host, and external storage as a hierarchy. Its radix metadata tracks local state, with backend lookup for external storage; matching, prefetch, and write-back are separate operations. The documented prefetch policies trade waiting for reuse against proceeding with available state [14]. A forecasting controller must compare against these existing policies, not against an engine with all offloading disabled.

CacheGen compresses KV representations and adapts context loading to network conditions [25]. Compression exchanges encoding/decoding and possible quality costs for fewer bytes. The relevant break-even expression becomes $t_{\text{encode}} + M_{\text{compressed}}/B + t_{\text{decode}}$, with any quality restriction treated as a constraint. It is not enough to report the compression ratio.

CacheBlend addresses reusable passages that are not an identical prefix by selectively recomputing part of the cached state to account for their new context [26]. This is more involved than concatenating cached arrays. Changing preceding tokens changes causal representations. A serving experiment using such reuse must state the method and test output quality, instead of assuming ordinary exact-prefix semantics.

Tutti, a 2026 preprint, explores a GPU-centric object/IO path for SSD-backed KV restoration, reducing CPU involvement in critical IO operations [36]. AsymCache, another 2026 preprint, links eviction and partial-context reuse to attention-kernel costs rather than maximizing cache hit rate alone [37]. Their relevance here is architectural: transfer orchestration and the location of missing tokens can matter as much as the count of missing bytes. Their reported speedups are not reproduced in this paper.

4.4 Prefill/decode disaggregation

A colocated engine executes both phases on the same worker group. A disaggregated system uses separate prefill and decode pools and transfers the required state between them. DistServe motivates this separation through independent phase SLOs, resource allocation, and parallelism plans, while accounting for transfer bandwidth [18].

The benefit is conditional. Let $\Delta t_{\text{interference}}$ be the colocation penalty removed and $\Delta t_{\text{specialization}}$ the gain from better phase-specific allocation. Disaggregation helps a latency objective only when those gains exceed transfer, extra queueing, and orchestration costs for the relevant request distribution. It can reduce tail ITL while leaving aggregate tokens/s unchanged or lower. The versioned vLLM disaggregation guide frames the feature around separate TTFT/ITL tuning and connector-based transfer, and labels it experimental in that version [12].

A necessary average transfer-capacity condition is

$$\lambda_{pd} \mathbb{E}[M_{pd}] < B_{pd, \text{usable}}, \quad (18)$$

where λ_{pd} is transferred requests per second. Even below this bound, burstiness and concurrent traffic can produce long transfer queues. With several links and routes, enforce the condition per bottleneck resource; adding all link bandwidths is invalid when transfers share a narrow hop. More prefill GPUs do not fix a saturated transfer fabric.

Mooncake’s KV-centric architecture combines phase separation with a distributed cache using other cluster memory/storage resources, and considers overload handling rather than assuming every arrival can be served [19]. This is directly relevant to ATFM: cache, queueing, and admission already interact in prior systems. The proposed distinction must be the information used and the validated decision improvement.

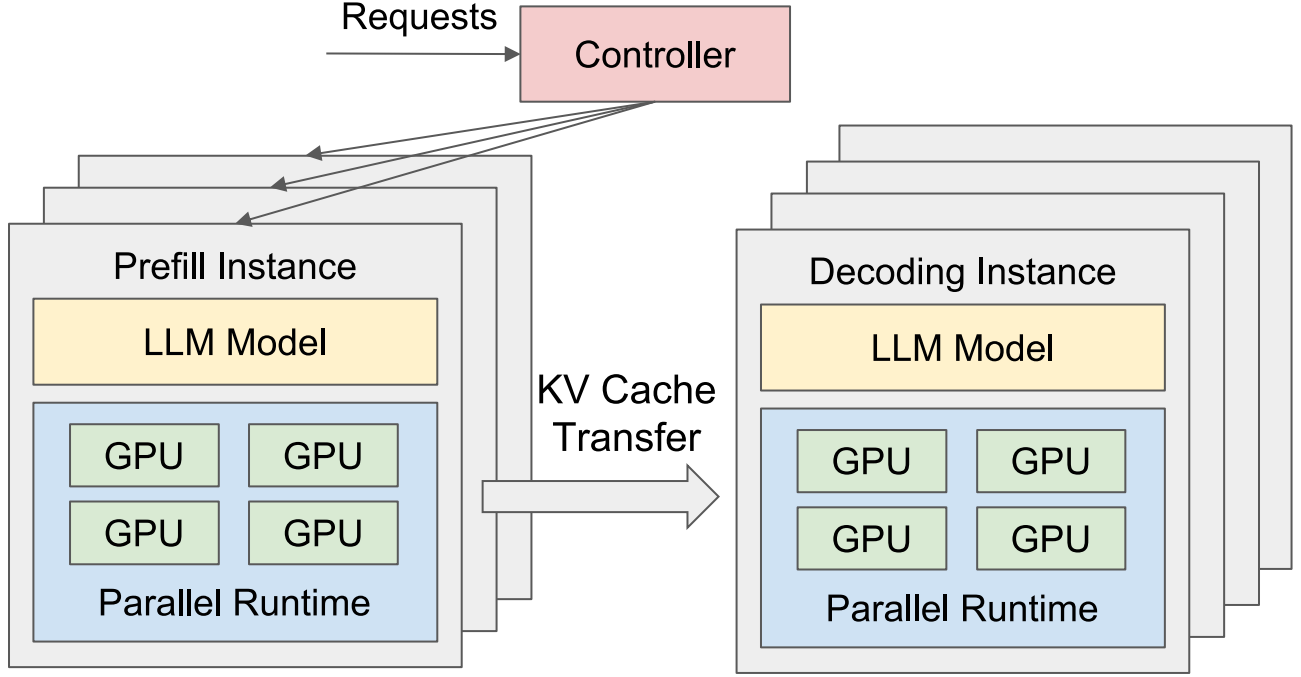


Figure 7: DistServe’s runtime architecture. The global scheduler coordinates separate prefill and decode worker groups; KV transfer connects their execution. Reproduced from Zhong et al. [18], Figure 6, arXiv:2401.09670v3, CC BY-SA 4.0; scaled and format-converted only.

4.5 Routing: reuse versus load

Routing all related prompts to one worker maximizes locality but may create a hot spot. Sending them to the shortest queue may discard valuable state. A conceptual score for destination j is

$$C_j = \hat{q}_j + \hat{t}_{\text{missing prefill},j} + \hat{t}_{\text{restore},j} + \hat{t}_{\text{decode},j} + \hat{c}_{\text{risk},j}. \quad (19)$$

All terms must have consistent units or explicit objective weights. Cache-hit fraction by itself is not a completion-time score. For example, a 95% hit with one second of queueing can lose to a 50% hit with 100 ms of extra prefill on an idle worker. This is a constructed illustration, not a measured threshold. Preble studies the joint optimization of distributed prompt reuse and load balancing [20].

NVIDIA Dynamo separates frontend handling, worker routing, execution, event feedback, and scaling/actuation. Its architecture includes backend-specific transfer coordination [30]. Its KV-aware routing documentation explicitly combines reusable cache state with projected active load, and separates selecting workers from moving their tensors [31]. Its planner already supports load-based and predictive scaling paths with aggregate traffic predictors [32]. ATFM cannot claim that serving systems only react or never forecast.

llm-d offers a Kubernetes-oriented composition of a gateway proxy, Endpoint Picker, inference pools, and model servers. The documented architecture includes cache-aware routing, cache indexing/offload, and disaggregated serving [33]. It is a deployment/control framework, whereas vLLM is an engine and LMCache is a cache layer. These components can compose; they are not three mutually exclusive answers to the same question.

4.6 State correctness and operational costs

KV state is derived but sensitive. It can encode information from user input, so cache sharing needs tenant isolation and authorization in addition to a collision-resistant identity scheme. A stale location record may

cause a miss; a wrong identity may cause an incorrect answer. Those failure modes deserve different treatment. Reference counts, leases, transfer completion, and invalidation must agree with engine ownership.

The costs of a cache policy include background writes, duplicate replicas, reserved host memory, network congestion, and abandoned prefetched data. A cache hit-rate improvement can be economically negative if it costs more resources than it saves. Similarly, a “touch” request that refreshes recency uses frontend and engine capacity and can perturb scheduling. It is a heuristic actuator, not a free, universal pin primitive.

Practical consequence. The decision is not simply whether KV exists. It is whether compatible state can be made ready at the right worker, before it is needed, at an acceptable cost to other work. ATFM’s forecasts would have to improve that decision beyond native cache and routing mechanisms.

5 Why agents change the scheduling problem

5.1 A session is a sequence of dependent visits

For session i , let turn k contain queueing Q_{ik} , inference I_{ik} , tool time Z_{ik} , and other orchestration O_{ik} . A sequential job has

$$J_i = \sum_k (Q_{ik} + I_{ik} + Z_{ik} + O_{ik}). \quad (20)$$

Parallel tools replace some sums with critical-path expressions. A fast individual LLM call does not guarantee fast J_i , and a policy that improves mean request latency can still harm long multi-turn jobs. Conversely, if tools dominate and inference is already fast, substantial GPU acceleration may barely change completion time.

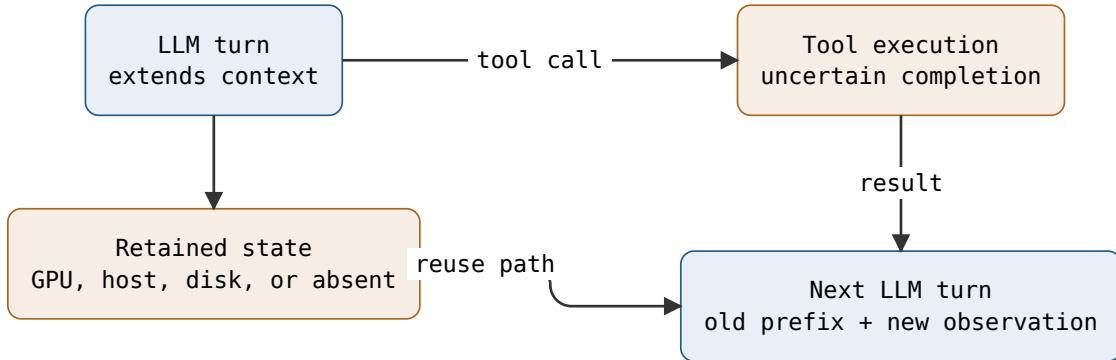


Figure 8: One cycle of an agent session. The tool gap is both a source of uncertainty and an opportunity to prepare state. An unchanged old prefix may remain reusable when a new observation is appended, subject to exact identity.

In a stationary closed population of N sequential sessions with mean inference visit response R and non-inference think/tool time Z , the interactive response-time relation is $X = N/(R + Z)$ visits per second. This follows by counting one completed cycle per session per average cycle duration. Increasing engine speed reduces R and can increase the rate of future arrivals. Replaying fixed request timestamps misses this feedback. Admission delays also shift future returns, so the workload is partly affected by the policy being evaluated.

AgentSysBench’s 2026 preprint studies ten agentic applications and reports heterogeneous bottlenecks, substantial non-LLM costs, and retained state across idle intervals [38]. This supports measuring complete

workflows; it does not prove that all agents have the same return-time distribution. MLCommons’ July 2026 agentic-inference announcement similarly emphasizes growing context and dependent multi-turn execution [39]. A benchmark proposal or announcement must not be confused with a completed, universally adopted performance standard.

5.2 Queueing: why small demand errors matter near saturation

For a deliberately simple M/M/1 system with Poisson arrivals, exponential service, one server, and utilization $\rho = \lambda/\mu < 1$,

$$\mathbb{E}[R] = \frac{1}{\mu - \lambda} = \frac{1/\mu}{1 - \rho}. \quad (21)$$

At utilization 0.5, mean response is twice mean service; at 0.9 it is ten times; at 0.99 it is one hundred times. A batched GPU system is not M/M/1, but this example explains why headroom and bursts matter. Mean capacity alone cannot predict tail latency when the service process changes with batching and state.

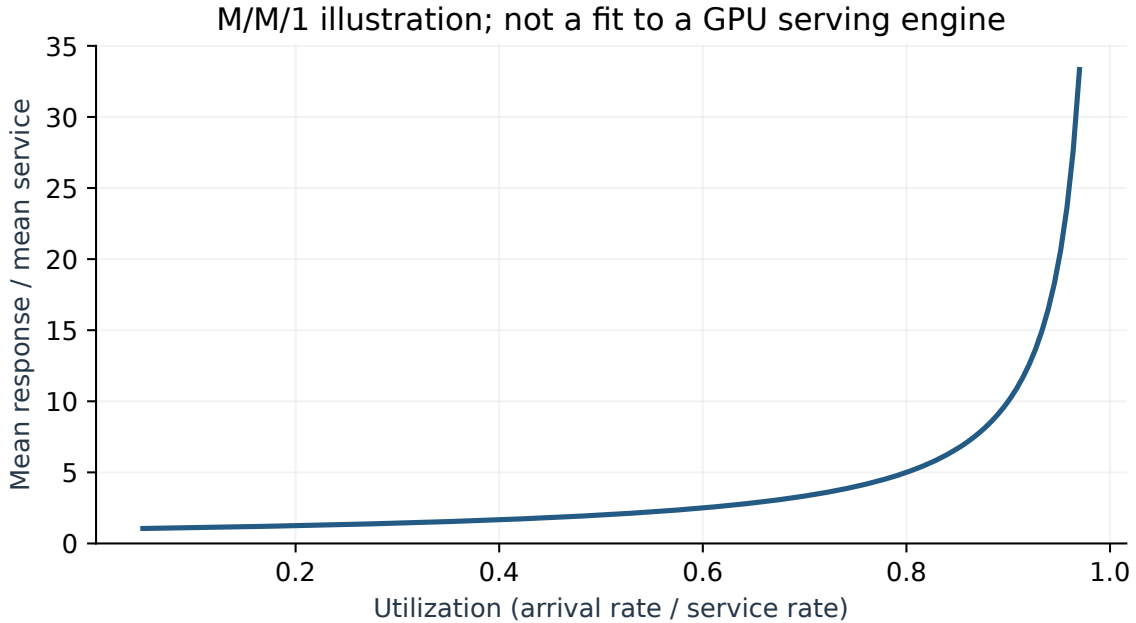


Figure 9: Analytical queueing illustration. The steep rise near saturation motivates controlling overload; it is not an empirical fit to ATFM, vLLM, or a real GPU.

If a resource has capacity C_r work units per second and arrival work demand $\lambda \mathbb{E}[D_r]$, a necessary long-run stability condition is $\lambda \mathbb{E}[D_r] < C_r$. The resource may be compute, link bytes, CPU time, or storage IO. Memory is also a stock constraint: instantaneous occupancy must fit, not just average allocation rate. Holding background work can shift peaks; it cannot make permanently excessive admitted work disappear. Rejection, additional capacity, or a lower work budget is eventually necessary.

A useful policy must state whose latency it improves. Giving all interactive traffic priority can starve background jobs. A finite delay cap protects background progress but may force release into an already overloaded system. Thus bounded delay and unconditional overload avoidance cannot both be promised under arbitrary arrivals. This is a policy tradeoff, not an implementation bug.

5.3 Forecasting the remaining tool time

Let T be a tool’s total duration, a its elapsed age, and $\mathcal{I}_t$ the information available now. With survival function $S(x) = \mathbb{P}(T > x)$, the age-conditioned residual distribution is

$$\mathbb{P}(T - a \leq h \mid T > a) = 1 - \frac{S(a + h)}{S(a)}, \quad S(a) > 0. \quad (22)$$

A tool that has already run for 60 seconds is not a fresh draw from its original duration distribution. Heavy tails can make expected remaining time increase with age; the memoryless exponential is a special case. Completed-only training can also bias estimates when long-running calls are censored by observation windows or cancellation.

Progress can provide additional evidence. If a tool reports completed work c out of total w at uncertain rate V , a simple remaining-time model is

$$R = \frac{w - c}{V} + R_{\text{finalize}}. \quad (23)$$

This is a model, not a universal parser truth. Work units may change cost over time; setup/finalization may dominate; multiple stages can reset counters; a reported percentage can be nonlinear. A robust system should retain uncertainty and detect invalid or stale progress rather than extrapolate one point as a certain completion timestamp.

For a concrete illustration, two tools have both run 20 seconds. One reports 950 of 1000 uniform items completed at a stable 50 items/s; the other reports 100 of 1000 at 5 items/s. The naive residual point estimates are one second and 180 seconds before finalization. Age alone cannot distinguish them. Whether this extra distinction is actionable depends on cache size, transfer lead time, and forecast reliability.

5.4 From return times to fleet demand

Let R_i be the next return time relative to now and D_{ir} the resource demand of that return. A simplified cumulative demand process is

$$A_r(h) = \sum_{i=1}^N D_{ir} \mathbf{1}\{R_i \leq h\} + A_{r,\text{new}}(h). \quad (24)$$

An actual agent can return several times within a long horizon, requiring a multi-visit model. The one-return form makes the distinction between timing and size explicit. Forecasting only request count misses long prompts and large contexts; forecasting only mean bytes misses synchronized return bursts.

In general,

$$\mathbb{E}[D_{ir} \mathbf{1}\{R_i \leq h\}] \neq \mathbb{E}[D_{ir}] \mathbb{P}(R_i \leq h) \quad (25)$$

unless the relevant conditional independence holds. Slow tools can produce larger observations. Backend congestion can correlate many sessions’ durations. Monte Carlo samples should preserve such dependencies when supported by data. For indicator returns with equal demand d ,

$$\text{Var}(A) = d^2 \left[\sum_i p_i (1 - p_i) + 2 \sum_{i < j} \text{Cov}(\mathbf{1}_i, \mathbf{1}_j) \right]. \quad (26)$$

Assuming independence deletes the covariance term and can understate tail risk. A forecast should be evaluated under shared backend stalls and release bursts, not only independently overlaid sessions.

Cumulative arriving KV demand is not instantaneous resident memory. Occupancy also depends on completion, eviction, sharing, output growth, and placement. Converting arrival quantiles into memory safety therefore needs a state/lifetime model or a deliberately conservative bound. Similarly, a prefill-token capacity per time slot is a calibrated work proxy, not a hardware invariant.

5.5 Retention as a decision under uncertainty

Suppose keeping one session’s state until either it returns or timeout τ avoids miss cost C if it returns in time. Let F be its residual return CDF, m bytes occupied, and λ_m the opportunity cost per byte-second in the same objective units as C . Relative to immediate eviction, a simple expected net value is

$$V(\tau) = CF(\tau) - \lambda_m m \mathbb{E}[\min(R, \tau)] = CF(\tau) - \lambda_m m \int_0^\tau (1 - F(u)) du. \quad (27)$$

This assumes constant miss benefit and memory price and ignores interactions, shared blocks, and capacity discontinuities. It is a useful derivation, not the algorithm implemented in ATFM. With density f ,

$$V'(\tau) = Cf(\tau) - \lambda_m m(1 - F(\tau)). \quad (28)$$

Where survival is positive, extending retention is locally favorable when $Ch(\tau) > \lambda_m m$, with hazard $h = f/(1-F)$. A point estimate of return time loses the uncertainty needed for this calculation. Multiple local optima are possible; a local hazard test alone is not a global optimization method.

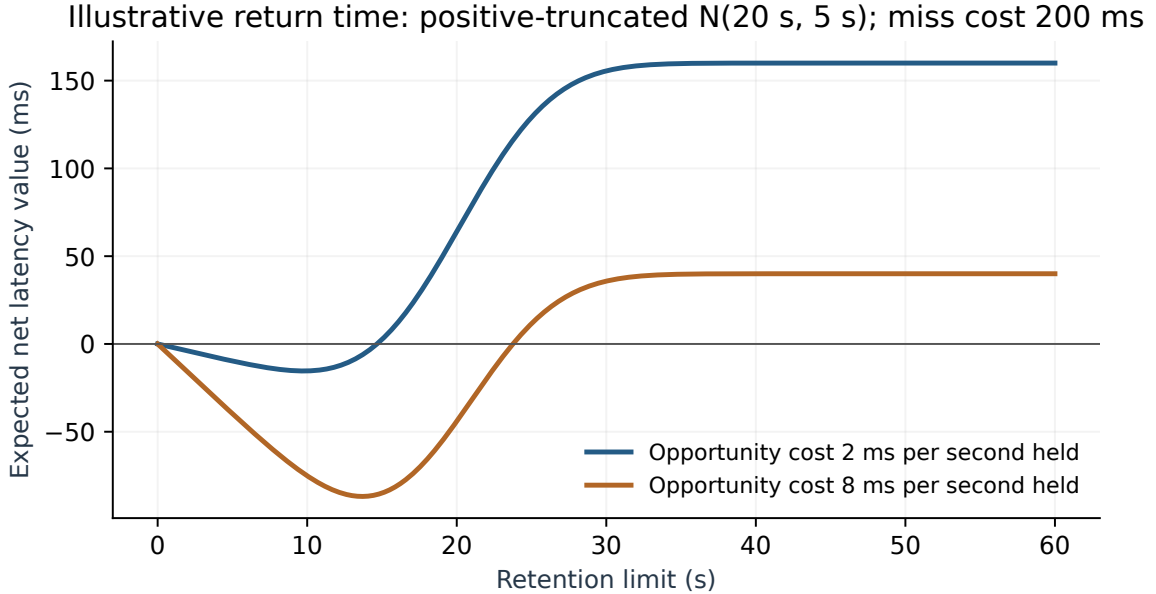


Figure 10: Derived retention value under two assumed memory opportunity costs. The return distribution is a normal distribution truncated to positive time. The miss penalty is 200 ms. The illustration shows that a forecast alone does not determine a retention policy; congestion changes the value of memory.

Prefetch has a related timing constraint. If transfer requires lead time ℓ and starts s seconds from now, then its probability of completing before reuse is approximately $\mathbb{P}(R \geq s + \ell)$ under a fixed-lead-time model. Starting too early occupies scarce space and can evict useful state. Starting too late leaves a demand-path stall. Transfer times themselves can be random and correlated with the same bursts the policy is trying to manage.

5.6 The closest prior work

Continuum’s inspected September 2026 revision uses bounded KV retention with a time-to-live mechanism and program-level scheduling. Its motivation includes reload/recompute costs, queueing after eviction, and robustness to uncertain tool duration [34]. It is a direct baseline for a claim about agent cache retention, not

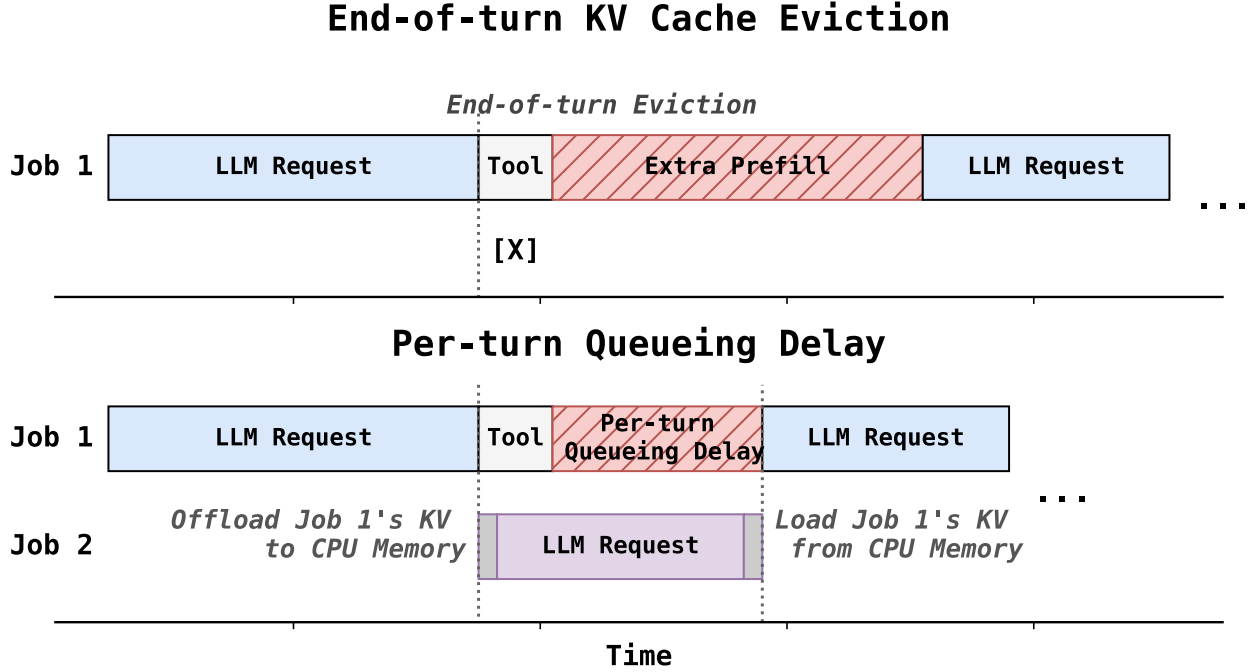


Figure 11: Continuum’s original cache-retention motivation. Compare the illustrated agent execution and cache lifetimes when reading its TTL policy. Reproduced from Li et al. [34], Figure 1, arXiv:2511.02230v7, CC BY 4.0; scaled and format-converted only.

merely background context. An ATFM comparison should use the actual supported policy or label a simplified reproduction honestly.

ThunderAgent’s June 2026 revision represents agent workflows as programs and coordinates KV-related scheduling with tool resources and environment preparation [35]. It already addresses the mismatch between isolated request decisions and workflow needs. ATFM’s possible contribution is a particular use of conditional forecasts and cross-session uncertainty, not the idea of treating agents as programs.

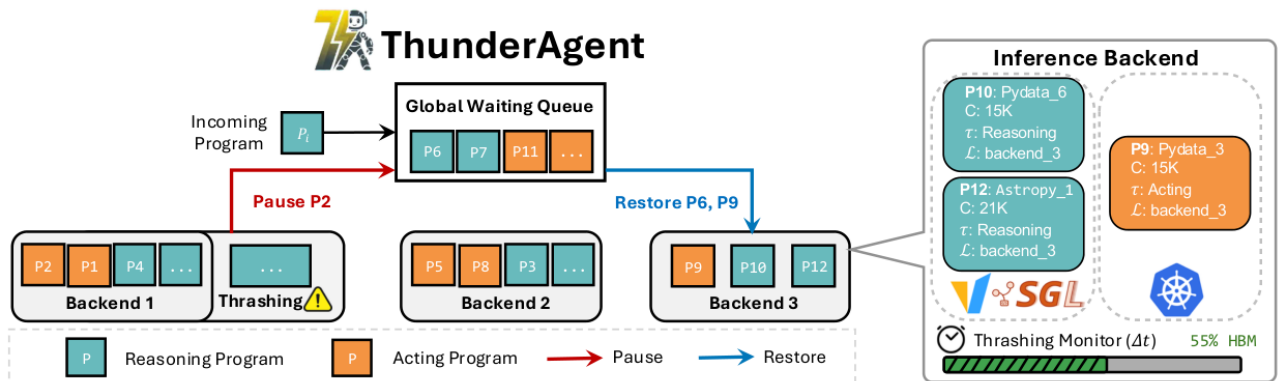


Figure 12: ThunderAgent’s original system overview, showing program-aware coordination of inference and tool execution. Reproduced from Kang et al. [35], Figure 3, arXiv:2602.13692v3, CC BY 4.0; scaled only.

These systems also bound the research claim. If a simple robust TTL, native retention priority, or working-set policy captures most of the benefit, a more complex predictor may not be worth its cost. The right experiment asks what extra information changes a decision, not whether the system beats an intentionally oblivious baseline.

6 Where ATFM fits, and what would make it useful

6.1 The specific information advantage being proposed

ATFM observes session/tool events, maintains a live forecast board, and turns forecasts into admission or placement-related directives. Its potential advantage is *lead time*: a tool-progress update may reveal a return before a gateway sees the next LLM request. A router choosing among destinations for an already-arrived prompt solves a different timing problem, although modern routing and scaling systems already use predictive models.

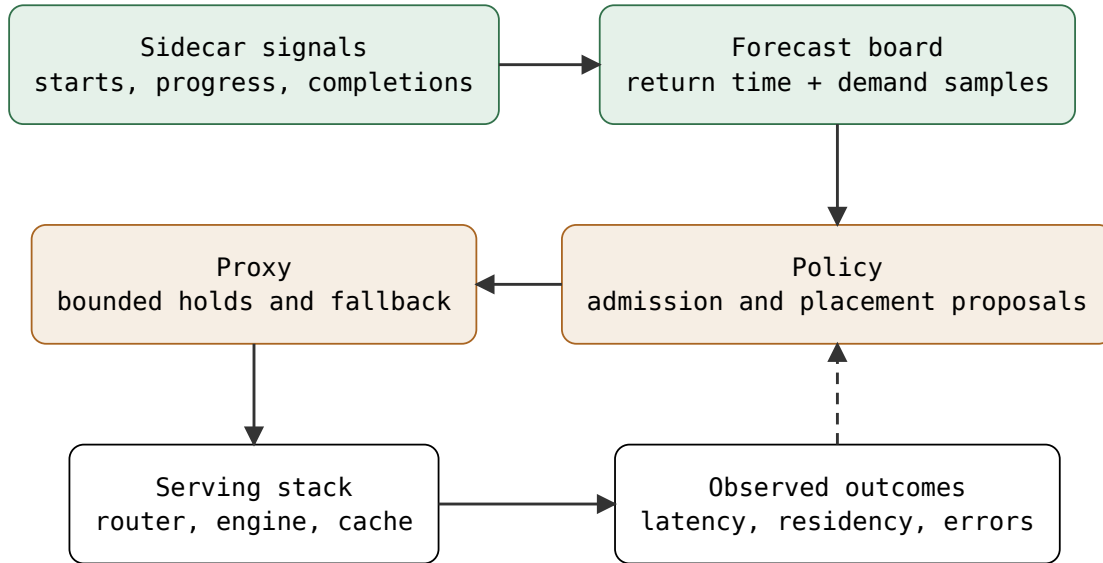


Figure 13: ATFM’s conceptual feedback loop. Forecasts become bounded control decisions; observed outcomes must validate that decisions were delivered and useful. Cache actuation is backend-dependent and is not depicted as a universal direct GPU pin.

The candidate contribution has three parts: conditional forecasts from session-specific observations; aggregate resource uncertainty rather than only mean request rate; and decisions made early enough to affect future contention. Each must earn its complexity. A high-quality forecast delivered after the request arrives has no prefetch lead-time advantage. A prediction that changes no action has no decision value. An action that increases another tenant’s latency more than it saves may be undesirable even when the forecast is correct.

The architecture suggests a useful separation. Keep the request path bounded and capable of falling back. Run heavier forecasting on snapshots or an independently budgeted path. Deliver versioned, expiring decisions to actuators that own the affected resource. This is a design direction; it must not be mistaken for a claim that every isolation and acknowledgement mechanism is already implemented.

6.2 Admission constraints: population risk and sample risk

For resource r in slot s , let Y_{rs} be uncertain protected demand, a_{is} a binary assignment of background job i , and d_{ir} its estimated demand. A conceptual marginal chance constraint is

$$\mathbb{P}\left(Y_{rs} + \sum_i a_{is}d_{ir} \leq C_{rs}\right) \geq 1 - \epsilon_{rs}. \quad (29)$$

Capacity C_{rs} must use the same units and time interpretation as demand. Existing resident KV and future arriving KV are not interchangeable. A system that imposes separate resource constraints does not automatically have a joint $1 - \epsilon$ guarantee across resources and slots. A conservative union bound allocates risks so $\sum_{r,s} \epsilon_{rs} \leq \epsilon_{\text{joint}}$; a joint-scenario method is another option.

With K forecast samples, a planner can evaluate

$$\frac{1}{K} \sum_{k=1}^K \mathbf{1}\left\{Y_{rs}^{(k)} + \sum_i a_{is}d_{ir} \leq C_{rs}\right\} \geq 1 - \epsilon_{rs}. \quad (30)$$

This is an empirical constraint under the forecast sample distribution. It is not automatically a frequentist guarantee for true future traffic. Finite sample uncertainty, model error, drift, and correlated residuals all matter. As a simple illustration, even for independent correctly generated draws, a Bernoulli event with probability 0.1 has estimated-probability standard error $\sqrt{0.1 \cdot 0.9/256} \approx 0.0188$ at 256 samples. Increasing samples reduces Monte Carlo noise but does not correct model bias.

ATFM's GDP planner assigns deferrable work in forecast order using sampled resource thresholds, bounded holds, and tenant-delay limits. Delay caps can release work even when the capacity test has no feasible slot. Consequently this implementation should be described as a bounded-delay heuristic with empirical feasibility checks, not a proof of overload prevention. The code and its scaling notes state the mechanics; a deployable capacity model remains an experimental responsibility.

6.3 A worked use case and its failure cases

Imagine a worker pool near a latency limit. Several background agents are running tests; an interactive user is generating a response. Some agents report that their tests are nearly complete and will soon return with long contexts. If this information is reliable and arrives early, a controller could defer new deferrable tool launches, preserve selected cache state, or prepare capacity. The objective is to reduce the coming peak, not to speed up the tests or invent more GPU capacity.

Now change the assumptions. If the tests emit misleading progress, the controller can hold unrelated work unnecessarily. If the contexts are small, prefetch/control overhead can exceed the saved prefill. If the engine has ample headroom, delaying anything may only increase completion time. If link bandwidth is saturated, prefetching the predicted burst can congest it earlier. If an existing TTL policy already retains the right prefixes, the extra forecast may be redundant. If tool time dominates each job, a per-call latency improvement may have negligible job-level impact.

A useful deployment region therefore requires all of the following to some degree: repeated state, scarce resources, exploitable advance information, a supported actuator, and a benefit greater than control/interference costs. The size of that region is an empirical question. It cannot be established by showing that one regression predicts durations accurately.

6.4 Comparison by responsibility, not headline speedup

Table 2: Relevant systems and the comparison ATFM needs. Descriptions refer to cited mechanisms, not a claim that other systems cannot be extended.

System	Established focus	What an ATFM comparison must isolate
vLLM	Engine scheduling, paged state, prefix reuse	Incremental benefit on a tuned engine with native reuse enabled
SGLang / HiCache	Program execution, radix reuse, cache tiers	Extra value over existing match/prefetch/eviction policies
TensorRT-LLM	Optimized execution, reuse and retention controls	Prediction benefit beyond available native priorities
LMCache	State retention, movement, cross-engine reuse	Which supported operation a forecast improves, including its cost
Sarathi-Serve	Chunked prefill and phase interference	Admission benefit after reasonable intra-engine scheduling
DistServe	Separate prefill/decode resource allocation	Whether forecast value survives transfer and pool queueing
Mooncake	KV-centric distributed serving and overload control	Extra session information versus existing cluster policy
Preble	Distributed prefix reuse and load balancing	Anticipatory timing versus locality/load decisions at arrival
Dynamo	Routing, transfer coordination, scaling	Session-conditional signals beyond aggregate predictors
llm-d	Kubernetes inference routing and deployment composition	Integration ownership and comparison with enabled policies
Continuum	Bounded retention and program scheduling	Conditional forecasts versus robust TTL decisions
ThunderAgent	Program-aware inference and tool resources	Additional forecast value at equal workflow visibility
Tutti / AsymCache	IO- or kernel-aware cache optimization	Complementarity with a more efficient cache data path
ATFM prototype	Tool signals, sampled forecasts, bounded directives	End-to-end benefit remains to be established on real workers

6.5 What the repository currently demonstrates

This assessment is anchored to the repository at 900d63b, not an assumption that documentation implies hardware validation. Local source paths are listed in Appendix E.

- Offline trace adapters, session forecasting, Monte Carlo aggregation, calibration, and evaluation are implemented. Recorded H1/H1b findings depend on corpus, split, horizon, and signal quality.
- The simulator contains queues, modeled workers, cache eviction, admission, and placement-related comparisons. It permits controlled counterfactuals but is not a real inference engine.
- The board, proxy, event ingestion, and directive paths have local HTTP and mock-worker integration evidence. A recorded Mocker setup lacked required KV capacity metrics and issued no holds or touches; that run verifies wiring, not the efficacy of those controls.
- The LMCache HTTP adapter is tested against controlled responses. Current controller compatibility,

completed transfers, and actual GPU cache effects remain unverified on the target deployment.

- Tier and replica-floor proposals can be logged; the default launcher is not evidence of a fully connected production autoscaler.
- The completed refactor passed 420 tests with five skips. Deterministic policy and forecast fixtures retained output hashes. This supports code maintenance, not a GPU goodput or model-quality claim.

The status document also states that a completed H100 performance study is not part of the recorded local evidence. The strongest useful present claim is that ATFM provides a testable forecasting/control prototype with measured CPU and integration behavior. Production serving advantage is still a hypothesis.

6.6 Known control costs and the Python/Rust decision

The repository has already removed repeated full-history JSONL scans, batched hold delivery, introduced persistent admission heaps, and reduced planner and empirical-sampling overhead. These changes target work volume before changing implementation language. The following bounds distinguish independent scale variables; there is no single meaningful N for the whole system.

Table 3: Selected ATFM costs after the documented control refactors. Bounds omit model-specific setup and external IO unless named.

Operation	Principal scaling	Remaining qualification
Incremental log drain	$O(\Delta E)$ newly read bytes/records	First read and restart still replay retained history
Heap admission updates	Amortized $O(\log Q)$ per heap operation	Due timers and compaction can cause bursts
Peer rank diagnostics	$O(Q)$ collection	Still appears in the HTTP priority path
Batched hold transport	$\lceil D/B \rceil$ RPCs, $O(D)$ payload	No whole-plan distributed transaction
Monte Carlo aggregation	Depends on N, K, H ; typically at least $O(NK)$ sample work	Vectorization does not eliminate samples or output arrays
GDP slot search	Range pruning; worst-case linear in eligible slots per job	Resource minima can occur in different slots

Here ΔE is newly appended event data, Q queued requests, D directives, B directives per transport batch, N sessions, K samples, and H horizons. Worst-case GDP work can be $O(NF)$ for F eligible slots; if both grow together, this is quadratic. Additional sample and resource dimensions belong explicitly in threshold construction. An index does not guarantee a logarithmic search when its pruning condition is only necessary. By contrast, returning a full Q -element diagnostic snapshot necessarily costs $\Omega(Q)$.

Recent local proxy profiling found repeated runtime-detection import work and removed it by supplying a missing serving dependency. Its paired fake-worker trials improved latency/CPU in that configuration, but prediction timeout rates increased. That result is recorded in the proxy-profiling report. It demonstrates why a lower CPU total is not sufficient evidence of a healthier control loop. The September 28 implementation bounds unfinished proxy prediction jobs (four by default), falls back immediately at capacity, and keeps abandoned running jobs counted until completion. A monotonic deadline spans dispatch and execution; board HTTP phases use its remaining budget instead of a 500 ms minimum. Caller outcomes and actual unfinished work are counted separately. This bounds local work, not remote computation or hard real-time response. It can exchange timeouts for immediate rejection, so timely prediction use must be measured alongside request latency. Synchronous board computation and remaining linear rank work are still investigation targets.

Python is currently appropriate for orchestration and numerical experimentation, with bulk numerical work handled by compiled libraries. Rust becomes attractive for a *measured* high-frequency CPU component when

algorithmic fixes, batching, and boundary costs have already been addressed. A native module should have a narrow contract and differential tests; moving the whole system would add maintenance cost and could leave the real bottleneck unchanged. Transferring large KV tensors and scheduling small metadata are separate costs, so neither “Python is always negligible” nor “Rust will fix saturation” is justified.

6.7 An evaluation that could validate or reject the project

Stage 1: establish executable contracts. Pin model, tokenizer/template, engine, cache connector, transport, and GPU configuration. Verify the cache identity, supported retention/movement operation, completion acknowledgement, expiration behavior, and observed residency. Record whether the action protects GPU, CPU, disk, or a remote copy. A mock endpoint returning 200 is insufficient. Test disconnects, stale events, partial failures, and fallback; measure whether a retry changes the same object or accidentally creates duplicate work.

Stage 2: evaluate forecasts without leakage. Split by complete session or workflow family and time, not arbitrary events from the same run. Report residual-time calibration and quantile loss at operational horizons. Compare aggregate last-value/rate predictors, age-only survival, tool-type models, and progress-conditioned models. Include missing progress, censored calls, unseen tools, and correlated backend slowdowns. A forecaster may be well calibrated overall and systematically wrong for the largest, most expensive contexts; stratify accordingly.

For quantile q_α and outcome y , pinball loss is

$$L_\alpha(y, q_\alpha) = (\alpha - \mathbf{1}\{y < q_\alpha\})(y - q_\alpha). \quad (31)$$

Report empirical interval coverage as well as width. Narrow incorrect intervals are not better forecasts. Evaluate service demand jointly with timing when they are dependent, and retain deterministic seeds for parity tests without treating seeds as independent production workloads.

Stage 3: isolate decision value. Use the same tuned engine and cache configuration across arms: native policy; robust TTL/working-set baseline; aggregate forecasting; session age-only forecasting; progress-conditioned ATFM; and an oracle whose extra information is explicit. Disable individual actuators to separate admission, retention, prefetch, and scaling effects. A simulated “ThunderAgent-style” policy must not be labelled a benchmark of the full ThunderAgent implementation. Likewise, a clairvoyant cache schedule is an upper bound under its modeled constraints, not a deployable competitor.

Stage 4: run closed-loop agent workloads on real workers. Vary sessions, prompt/output lengths, tool gaps, shared-prefix fraction, cache capacity, transfer bandwidth, burst correlation, tenant mix, and forecast error. Include low load where control can only add overhead, near-saturation load where it may help, and sustained overload where admission must make tradeoffs. Use both reproducible synthetic programs and real tool traces. Preserve causal turn dependencies so latency changes alter future arrival times correctly.

Stage 5: account for the entire outcome. Measure offered/completed/rejected work, joint-SLO goodput, TTFT, ITL tails, whole-job completion, fairness, cache bytes by tier, useful/wasted transfers, recomputed tokens, GPU idle time, controller CPU, queueing, and prediction deadline misses. Include warmup, shutdown/drain rules, retries, errors, and load-generator lateness. Run paired workloads with alternating order and confidence intervals at the independent workload/run level. Correlated tokens from one request are not independent experimental replicates.

Success and falsification criteria. A positive result should show an improved user-relevant objective at equal resource budget, no unreported quality loss, and acceptable fairness/overhead. A negative result is informative if simple TTL or native policies match the benefit, useful lead time is scarce, prediction deadlines are missed, or cache movement dominates savings. The project remains useful only if the value of extra information survives the full path from observation to completed action.

A Notation and units

Symbol	Meaning
L, d, d_h	Layer count, hidden width, per-head dimension
H_q, H_{kv}	Query heads and KV heads; they need not be equal
T, U	Prompt/context length and number of generated output tokens
b	Bytes per stored scalar; scales/metadata may be additional
B	Active batch size in engine examples; directive batch size only in the explicitly defined ATFM table
p	Tokens per physical cache block
M_{KV}	KV payload bytes, with sharding/replication specified separately
P_{eff}	Effective compute rate for the actual operations/shapes
$B_{\text{effective}}$	Effective bandwidth in bytes/s, not bits/s
Q, N, K, H, F	Queued requests, sessions, Monte Carlo samples, forecast horizons, and candidate time slots
R_i, D_{ir}	Session return time and its demand for resource r
C_{rs}	Usable capacity for resource r in slot s under a stated model
$F(t), S(t), h(t)$	CDF, survival function, and hazard of a duration/residual distribution
ρ	Utilization under the simplified queue model
τ, ϵ	A time/SLO bound and a probability-risk budget, respectively

A KiB is 2^{10} bytes and a GiB is 2^{30} bytes; a GB is 10^9 bytes. A 100 Gb/s link has a raw decimal byte rate of 12.5 GB/s before protocol and implementation overhead, approximately 11.64 GiB/s. Do not substitute 100 GB/s into a transfer equation for such a link. In the illustrations, bandwidth is explicitly in GiB/s to match payload units. A model’s advertised maximum context length is a supported per-sequence limit, not a promise that many maximum-length sequences fit or meet an SLO simultaneously.

B Terminology that is easy to conflate

KV cache versus response cache

KV caches intermediate attention state. Response caching returns a previously produced answer and requires its own semantic/freshness contract. Reusing KV can still produce a different output sample.

Prefix reuse versus session identity

A session label is application metadata. Exact reuse depends on compatible model state and prefix tokens. Two different sessions may share a prefix; two requests in one session may lose reuse after editing earlier context.

Cache hit versus ready state

A metadata hit says an object is known at a location. Execution also requires valid ownership, complete transfer, compatible layout, and synchronization.

Eviction versus offload versus deletion

Eviction removes an object from a tier's available resident set. Offload preserves a representation elsewhere. Deletion removes the retained representation from the relevant store. Exact meanings are API-specific.

Pin versus priority versus touch

A pin normally constrains eviction in a defined location. Priority changes relative preference. A touch causes an access that may affect recency. They are not interchangeable guarantees, and an external-store pin need not pin engine HBM.

Continuous batching versus request concurrency

Concurrency counts outstanding requests. Continuous batching chooses token work between iterations. Large concurrency can increase queueing without increasing useful device work.

Prefill/decode disaggregation versus model parallelism

Disaggregation separates request phases. Model parallelism splits one phase's model execution across devices. They can be combined.

Goodput versus throughput

Throughput counts completed work under a chosen unit. Goodput counts work meeting an explicit quality or latency condition. Both need the offered workload and resource budget for interpretation.

Prediction interval versus capacity guarantee

A calibrated interval describes outcome frequency under an evaluated distribution. It is not proof that a controller remains safe after arbitrary distribution shift or that joint resource risks have been bounded.

C A complete example: one agent returning to a changed fleet

This example is constructed for explanation. It is not a simulator result or a measurement. Use the 128 KiB/token cache geometry from Equation 5. An agent has an 8192-token prefix (1 GiB), finishes a turn, and launches a tool. Another workload needs memory while it waits.

1. **At departure**, the engine releases active references according to its policy. Retained blocks may remain reusable but evictable. An external cache may also have a copy; do not assume one was written merely because offload is configured.
2. **During the gap**, a progress update changes the predicted remaining time to approximately two seconds with uncertainty. If a CPU copy exists, a forecasting policy might arrange a restore shortly before the expected return. It must include current transfer queueing and destination capacity.
3. **At arrival**, the tool result adds 512 tokens. A perfect old-prefix hit still requires processing that suffix and its attention to prior context. The full new payload at 8704 tokens is 1.0625 GiB, before output growth.
4. **At routing**, worker A has the old prefix but 200 ms of estimated queueing. Worker B is idle but would pay an assumed 150 ms recomputation penalty. Ignoring all other differences, B can win. If A's queue is only 20 ms, A can win. Locality must be valued in time, not treated as an absolute preference.
5. **After a restore**, an operation acknowledgement must establish which bytes became usable and where. If it only confirms admission to an asynchronous transfer queue, the engine may still be unable to start.
6. **At evaluation**, compare job completion and other tenants' latency. Saving this agent 150 ms while delaying ten others by 30 ms each may worsen a sum-of-latencies objective. A class-weighted objective could prefer a different choice; state those weights explicitly.

A shared prefix further changes the accounting. If four sessions genuinely share the same 1 GiB physical prefix, its retained payload is not 4 GiB. It can also have higher reuse value than a private 1 GiB suffix. A session-level policy that ignores block sharing can double-count memory and independently evict or pin state whose ownership is shared. The actuator must reconcile these decisions with the engine’s object/block identities.

D A diagnosis guide for slow serving

Symptom	Candidate explanation	Measurement that distinguishes it
High TTFT, low GPU busy time	Queueing outside engine, CPU dispatch, cache IO	Per-stage trace; frontend queue age; CPU/event-loop profile; transfer completion
High TTFT, heavy prefill work	Long prompts or poor prefix reuse	Uncached input tokens; cache identity/hit lengths; prefill kernel duration
Long ITL spikes	Prefill interference, collectives, pauses	Iteration composition; chunk size; device trace; per-token delivery timing
Frequent recompute/preemption	Active-state pressure or weak retention	Free/used blocks; eviction reason; prefix age; memory by owner
High hit rate but little benefit	Remote hits, short saved prefixes, hot worker	Saved compute time versus lookup/restore/queue time
Throughput plateaus with spare GPU	CPU/router/generator saturation	Per-process CPU, event-loop lag, offered versus delivered arrival rate
Good call latency, poor job latency	Tools, orchestration, serial dependencies	Whole-session critical-path trace and task-success metric
Prediction fallbacks rise at scale	Shared CPU or unbounded work backlog	Prediction deadline misses, queue age, cancellation effectiveness
Prefetch improves one class only	Transfer/memory interference or unfairness	Per-tenant latency and bandwidth; wasted prefetch and retention time

Use a profiler to locate time and an uninstrumented run to measure the effect of a change. Instrumentation can distort short operations and asynchronous scheduling. A fake worker is useful for isolating HTTP and controller overhead, but cannot establish the balance of GPU kernels, HBM, collectives, and actual KV transfers. Likewise, an offline maximum-throughput benchmark is not evidence about an open arrival process under tail-latency constraints.

E Repository map and evidence trail

These paths are relative to the ATFM repository and identify the prototype’s implementation/evidence at the baseline revision. They are not upstream API contracts. The repository is available at <https://github.com/Alexander-Aghili/ATFM/tree/900d63b>.

Path	Responsibility or evidence
src/atfm/sidecar/	Tool execution, progress extraction, configuration, harness adapters
src/atfm/board/	Session state, forecasts, predictors, aggregation, live HTTP board

Path	Responsibility or evidence
<code>src/atfm/proxy/queue.py</code>	Persistent admission indexes and timed release
<code>src/atfm/control/gdp.py</code>	Bounded-delay slot assignment and empirical resource tests
<code>src/atfm/control/touch.py</code>	Budgeted touches and tier recommendations
<code>src/atfm/control/lmcache.py</code>	Configurable HTTP actuator; current upstream compatibility remains open
<code>src/atfm/sim/</code> <code>experiments/</code>	Modeled engine and policy comparisons CPU, serving, forecasting, and profiling studies outside the core package
<code>docs/status.md</code>	Implemented versus validated capabilities
<code>docs/development/control-scaling.md</code>	Algorithms, exact contracts, complexity, retained benchmarks
<code>docs/research/2026-09-27-proxy-profiling.md</code>	Proxy profiles, dependency correction, latency and timeout tradeoffs
<code>docs/research/2026-09-28-core-refactor.md</code>	Refactor decisions, parity hashes, timing limits, final tests
<code>docs/paper/control-implementation.tex</code>	Main research paper’s control implementation/evidence section

The background paper intentionally has its own source, figures, and bibliography. It can be rebuilt without rewriting the main research manuscript or its frozen results. Original architecture diagrams are editable reladraw files. Mathematical plots are generated by `plots.py`; `illustration-inputs.json` records assumptions. No illustration should be relabelled as experimental evidence.

F Reading sequence and open questions

A productive sequence is: attention and GQA for the representation; Orca and PagedAttention for scheduling and allocation; Sarathi-Serve and DistServe for phase interference; SGLang/HiCache, LMCache, and Mooncake for reuse and movement; Preble, Dynamo, and llm-d for fleet composition; then Continuum and ThunderAgent for the direct agent-scheduling comparison. Read the relevant current project documentation before implementing an integration: a paper explains a design, but does not pin today’s connector or controller API.

Three questions now connect this background to a falsifiable research plan: does progress improve forecasts at actionable lead times; can a real actuator convert those forecasts into ready-state advantages without displacing more valuable work; and does that advantage improve goodput or job completion over strong baselines after control costs and fairness are included?

References

- [1] Vaswani et al.. *Attention Is All You Need*. 2017. <https://arxiv.org/abs/1706.03762>. Accessed 28 September 2026.
- [2] Shazeer. *Fast Transformer Decoding: One Write-Head is All You Need*. 2019. <https://arxiv.org/abs/1911.02150>. Accessed 28 September 2026.
- [3] Ainslie et al.. *GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. 2023. <https://arxiv.org/abs/2305.13245>. Accessed 28 September 2026.
- [4] Dao et al.. *FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness*. 2022. <https://arxiv.org/abs/2205.14135>. Accessed 28 September 2026.
- [5] Shah et al.. *FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision*. 2024. <https://arxiv.org/abs/2407.08608>. Accessed 28 September 2026.
- [6] Ye et al.. *FlashInfer: Efficient and Customizable Attention Engine for LLM Inference Serving*. 2025. <https://arxiv.org/abs/2501.01005>. Accessed 28 September 2026.
- [7] Yu et al.. *Orca: A Distributed Serving System for Transformer-Based Generative Models*. 2022. <https://www.usenix.org/conference/osdi22/presentation/yu>. Accessed 28 September 2026.
- [8] Kwon et al.. *Efficient Memory Management for Large Language Model Serving with PagedAttention*. 2023. <https://arxiv.org/abs/2309.06180>. Accessed 28 September 2026.
- [9] vLLM contributors. *vLLM V1 guide*. living documentation. https://github.com/vllm-project/vllm/blob/main/docs/usage/v1_guide.md. Accessed 28 September 2026.
- [10] vLLM contributors. *Automatic Prefix Caching*. v0.15.0 documentation. https://docs.vllm.ai/en/v0.15.0/design/prefix_caching/. Accessed 28 September 2026.
- [11] vLLM contributors. *Hybrid KV Cache Manager*. living documentation. https://docs.vllm.ai/en/latest/design/hybrid_kv_cache_manager/. Accessed 28 September 2026.
- [12] vLLM contributors. *Disaggregated Prefilling (experimental)*. v0.20.0 documentation. https://docs.vllm.ai/en/v0.20.0/features/disagg_prefill/. Accessed 28 September 2026.
- [13] Zheng et al.. *SGLang: Efficient Execution of Structured Language Model Programs*. 2023. <https://arxiv.org/abs/2312.07104>. Accessed 28 September 2026.
- [14] SGLang contributors. *HiCache System Design and Optimization*. living documentation. https://docs.sglang.io/docs/advanced_features/hicache_design. Accessed 28 September 2026.
- [15] NVIDIA. *TensorRT-LLM KV Cache System*. living documentation. <https://nvidia.github.io/TensorRT-LLM/latest/features/kvcache.html>. Accessed 28 September 2026.
- [16] Prabhu et al.. *vAttention: Dynamic Memory Management for Serving LLMs without PagedAttention*. 2024. <https://arxiv.org/abs/2405.04437>. Accessed 28 September 2026.
- [17] Agrawal et al.. *Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve*. 2024. <https://arxiv.org/abs/2403.02310>. Accessed 28 September 2026.
- [18] Zhong et al.. *DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving*. 2024. <https://arxiv.org/abs/2401.09670v3>. Accessed 28 September 2026.
- [19] Qin et al.. *Mooncake: A KVCache-centric Disaggregated Architecture for LLM Serving*. 2025 revision. <https://arxiv.org/abs/2407.00079v4>. Accessed 28 September 2026.
- [20] Srivatsa et al.. *Preble: Efficient Distributed Prompt Scheduling for LLM Serving*. 2024. <https://arxiv.org/abs/2407.00023v2>. Accessed 28 September 2026.
- [21] Liu et al.. *LMCache: An Efficient KV Cache Layer for Enterprise-Scale LLM Inference*. 2025. <https://arxiv.org/abs/2510.09665v2>. Accessed 28 September 2026.

- [22] LMCache contributors. *Architecture and Developer Guide: multiprocess mode*. living documentation. <https://docs.lmcache.ai/mp/architecture.html>. Accessed 28 September 2026.
- [23] LMCache contributors. *LMCache Controller*. legacy documentation. https://docs.lmcache.ai/kv_cache_management/index.html. Accessed 28 September 2026.
- [24] LMCache contributors. *Pin the KV cache*. legacy documentation. https://docs.lmcache.ai/kv_cache_management/pin.html. Accessed 28 September 2026.
- [25] Liu et al.. *CacheGen: KV Cache Compression and Streaming for Fast Large Language Model Serving*. 2023. <https://arxiv.org/abs/2310.07240>. Accessed 28 September 2026.
- [26] Yao et al.. *CacheBlend: Fast Large Language Model Serving for RAG with Cached Knowledge Fusion*. 2025 revision. <https://arxiv.org/abs/2405.16444v3>. Accessed 28 September 2026.
- [27] Liu et al.. *KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache*. 2024. <https://arxiv.org/abs/2402.02750>. Accessed 28 September 2026.
- [28] DeepSeek-AI. *DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model*. 2024. <https://arxiv.org/abs/2405.04434>. Accessed 28 September 2026.
- [29] Leviathan et al.. *Fast Inference from Transformers via Speculative Decoding*. 2022. <https://arxiv.org/abs/2211.17192>. Accessed 28 September 2026.
- [30] NVIDIA. *Dynamo Architecture*. living documentation. <https://docs.nvidia.com/dynamo/dev/knowledge-base/concepts/architecture>. Accessed 28 September 2026.
- [31] NVIDIA. *Dynamo KV-Aware Routing*. living documentation. <https://docs.nvidia.com/dynamo/dev/knowledge-base/concepts/system-architecture/kv-aware-routing>. Accessed 28 September 2026.
- [32] NVIDIA. *Dynamo Planner*. living documentation. <https://docs.nvidia.com/dynamo/dev/knowledge-base/modular-components/planner/overview>. Accessed 28 September 2026.
- [33] llm-d contributors. *llm-d Architecture*. v0.9 documentation snapshot. <https://llm-d.ai/docs/architecture>. Accessed 28 September 2026.
- [34] Li et al.. *Continuum: Efficient and Robust Multi-Turn LLM Agent Scheduling with KV Cache Time-to-Live*. 2026 revision. <https://arxiv.org/abs/2511.02230v7>. Accessed 28 September 2026.
- [35] Kang et al.. *ThunderAgent: A Simple, Fast and Program-Aware Agentic Inference System*. 2026 revision. <https://arxiv.org/abs/2602.13692v3>. Accessed 28 September 2026.
- [36] Qiu et al.. *Tutti: Making SSD-Backed KV Cache Practical for Long-Context LLM Serving*. 2026 preprint. <https://arxiv.org/abs/2605.03375>. Accessed 28 September 2026.
- [37] Shi et al.. *Multi-Segment Attention: Enabling Efficient KV-Cache Management for Faster Large Language Model Serving*. 2026 preprint. <https://arxiv.org/abs/2606.02964>. Accessed 28 September 2026.
- [38] Chang et al.. *From LLM Inference to Agentic Workloads: Characterization and Implications for Serving Systems*. 2026 preprint. <https://arxiv.org/abs/2608.15127>. Accessed 28 September 2026.
- [39] MLCommons. *Agentic Inference for MLPerf Inference*. 8 July 2026. <https://mlcommons.org/2026/07/agentic-inference-for-mlperf-inference/>. Accessed 28 September 2026.